

ASSIGNMENT - 4.3

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Batch-10

Task-1

Prompt: Give me a program to zero-short prompt ng to check a leap year, and give instructions without providing examples

code :

```
def is_leap_year(year):    if (year % 4 == 0 and year % 100 !=
0) or (year % 400 == 0):
```

```
    return True    return False    year = int(input("Enter a
year to check if it's a leap year: ")) if is_leap_year(year):
    print(f"{year} is a leap year.")
```

else:

```
    print(f"{year} is not a leap year.")    Output
```

:

```
1 #Task-1
2 #Give me a program to zero-short prompting to check leap year and giving instructions without providing examples
3 def is_leap_year(year):
4     if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
5         return True
6     return False
7 year = int(input("Enter a year to check if it's a leap year: "))
8 if is_leap_year(year):
9     print(f"{year} is a leap year.")
10 else:
11     print(f"{year} is not a leap year.")
12 #~~~~~
13 #Task-2
14 #generate a one-short prompting that changes centimeters to inches with one input output with using correct math
```

PROBLEMS 13 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 13 Python + v

PS C:\Users\hp\OneDrive\Desktop\AI> & C:/Users/hp/AppData/Local/Microsoft/WindowsApps/python3.11.exe c:/Users/hp/OneDrive/De
.py
Enter a year to check if it's a leap year: 2024
2024 is a leap year.

Code Analysis:

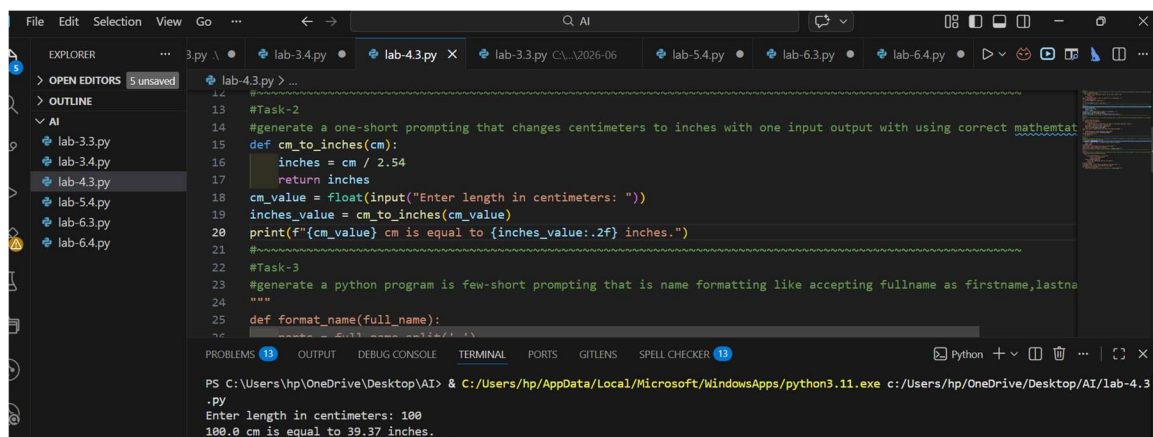
- ☐ This program determines whether a given year is a leap year using a function.
- ☐ The function applies standard leap year rules and returns True or False.
- ☐ The user inputs a year, which is checked by the function.
- ☐ The result is printed as either a leap year or not.

Task-2

Prompt: generate a one-short prompt that changes centimetres to inches with one input and output using the correct mathematical formula

Code :

```
def cm_to_inches(cm):  
    inches = cm / 2.54  
    return inches  
cm_value = float(input("Enter length  
in centimeters: "))  
inches_value =  
cm_to_inches(cm_value)  
print(f'{cm_value} cm  
is equal to {inches_value:.2f} inches.")
```

 Output :A screenshot of a code editor window. The editor has a dark theme. On the left, there's a sidebar with 'EXPLORER' and 'OUTLINE' tabs. The 'EXPLORER' tab shows a file explorer with several Python files. The 'OUTLINE' tab shows the structure of the code. The main editor area shows the Python code for Task 2. The code defines a function 'cm_to_inches' that takes a centimeter value and returns the equivalent in inches. It then prompts the user to enter a length in centimeters, calls the function, and prints the result. The terminal at the bottom shows the execution of the code, with the user entering '100' and the program outputting '100.0 cm is equal to 39.37 inches.'

Code Analysis:

- ☐ This program converts a length from centimetres to inches using the correct mathematical formula.
- ☐ A function performs the conversion by dividing the value by 2.54.
- ☐ The user enters a value in centimetres, which is passed to the function.
- ☐ The converted result is displayed in inches.

Task-3

Prompt: generate a python program is few-short prompt ng that is name forma ng like accep ng fullname as firstname,lastname.

Code :

```

def format_name(full_name):
    parts = full_name.split(',')
    if len(parts) != 2:
        raise ValueError("Please enter the name in 'Firstname,Lastname' format.")
    first_name = parts[0].strip().capitalize()
    last_name = parts[1].strip().capitalize()
    return f'{first_name} {last_name}'

full_name_input = input("Enter full name (Firstname,Lastname): ")

try:
    formatted_name = format_name(full_name_input)
    print(f'Formatted Name: {formatted_name}')
except ValueError as e:
    print(e)

```

Output:

The screenshot shows a VS Code editor with a Python file named 'lab-43.py'. The code defines a function 'format_name' that splits a full name into first and last names, capitalizes them, and returns the formatted string. It then prompts the user for input and prints the formatted name. The terminal output shows the user entering 'puvati,vamshi' and the program outputting 'Formatted Name: Puvati Vamshi'.

```

22 #Task-3
23 #generate a python program is few-short prompting that is name formatting like accepting fullname as firstname,lastname
24 def format_name(full_name):
25     parts = full_name.split(',')
26     if len(parts) != 2:
27         raise ValueError("Please enter the name in 'Firstname,Lastname' format.")
28     first_name = parts[0].strip().capitalize()
29     last_name = parts[1].strip().capitalize()
30     return f'{first_name} {last_name}'
31 full_name_input = input("Enter full name (Firstname,Lastname): ")
32 try:
33     formatted_name = format_name(full_name_input)
34     print(f'Formatted Name: {formatted_name}')
35 except ValueError as e:

```

Terminal Output:

```

PS C:\Users\hp\OneDrive\Desktop\AI> & C:/Users/hp/AppData/Local/Microsoft/WindowsApps/python3.11.exe c:/Users/hp/OneDrive/Desktop/AI/lab-43.py
Enter full name (Firstname,Lastname): puvati,vamshi
Formatted Name: Puvati Vamshi

```

Code Analysis :

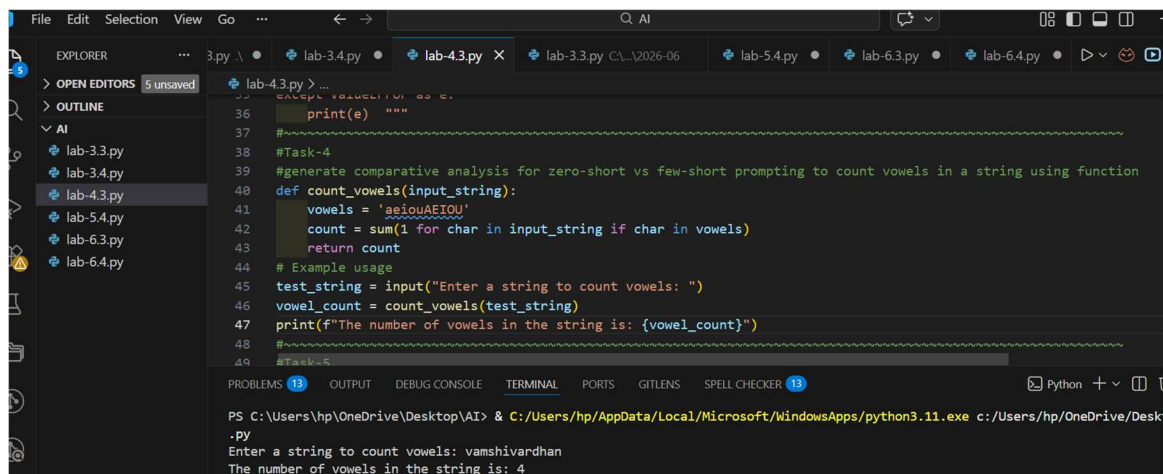
- ☐ This program formats a full name entered as first name and last name.
- ☐ The input is split and validated to ensure the correct format.
- ☐ Each part of the name is cleaned and capitalised. ☐ The formatted full name is then displayed.

Task-4

Prompt: generate a comparative analysis for zero-shot vs few-shot prompting to count vowels in a string using a function:

Code :

```
def count_vowels(input_string):  
    vowels = 'aeiouAEIOU'    count = sum(1 for char in input_string if  
    char in vowels)    return count  
  
# Example usage  
test_string = input("Enter a string to count vowels: ")  
  
vowel_count = count_vowels(test_string) print(f"The number  
of vowels in the string is: {vowel_count}") Output:
```



The screenshot shows a Visual Studio Code editor with a Python file named 'lab-4.3.py'. The code defines a function 'count_vowels' that takes an input string and returns the count of vowels. It includes an example usage section where a user is prompted to enter a string, and the output is printed. The terminal at the bottom shows the command 'python lab-4.3.py' being executed, with the input 'vamshivardhan' and the output 'The number of vowels in the string is: 4'.

Code Analysis :

- ☐ The function counts vowels in a given string using a direct logic approach.
- ☐ Zero-shot prompting applies the logic without examples.
- ☐ Few-shot prompting helps by showing patterns before execution.
- ☐ The function returns the total number of vowels in the input string.

Task-5

Prompt: generate a few short prompts for file handling to give a read text file, count the number of lines in the file, and line count by function

def count_lines_in_file(file_path):

try:

```

with open(file_path, 'r') as file:

    lines    =

file.readlines()    return

len(lines)    except

FileNotFoundError:

    print("The specified file was not found.")    return

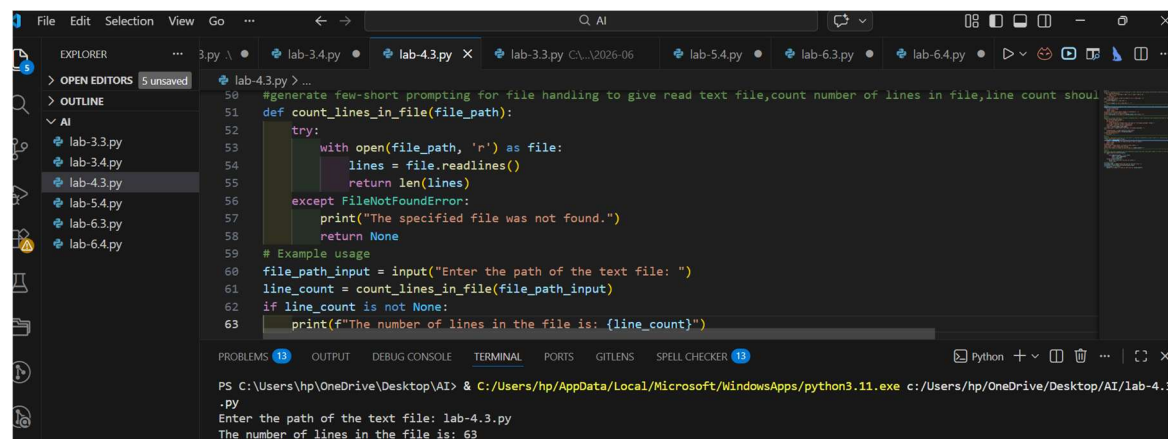
None

# Example usage
file_path_input = input("Enter the
path of the text file: ")
line_count =
count_lines_in_file(file_path_input)
if line_count is
not None:

    print(f"The number of lines in the file is: {line_count}")

```

Output :



The screenshot shows a Visual Studio Code editor with a Python file named 'lab-4.3.py' open. The code in the file is as follows:

```

50 #generate few-short prompting for file handling to give read text file,count number of lines in file,line count shoul
51 def count_lines_in_file(file_path):
52     try:
53         with open(file_path, 'r') as file:
54             lines = file.readlines()
55             return len(lines)
56     except FileNotFoundError:
57         print("The specified file was not found.")
58         return None
59 # Example usage
60 file_path_input = input("Enter the path of the text file: ")
61 line_count = count_lines_in_file(file_path_input)
62 if line_count is not None:
63     print(f"The number of lines in the file is: {line_count}")

```

The terminal at the bottom shows the command prompt where the file is executed:

```

PS C:\Users\hp\OneDrive\Desktop\AI> & C:/Users/hp/AppData/Local/Microsoft/WindowsApps/python3.11.exe c:/Users/hp/OneDrive/Desktop/AI/lab-4.3.py
Enter the path of the text file: lab-4.3.py
The number of lines in the file is: 63

```

Code Analysis :

- ☐ This program reads a text file and counts the number of lines.
- ☐ A function opens the file safely and calculates the line count.
- ☐ Error handling is used if the file does not exist.
- ☐ The final line count is returned and displayed.